NEURUH · AI THAT EXECUTES

Neuruh Capability Brief

Distributable subset of the Neuruh Codex. Component register, verified capability inventory, and commercial state, calibrated for external readers. Infrastructure detail and export-controlled content are excluded by design, not by omission.

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EVIDENCE Code-Verified Reality audit, 30 Jun 2026, reconciled against live state, 10 Nov 2026

Valuation Honesty Rule. This document contains no dollar valuation, no revenue projection, and no forward financial claim. Neuruh operates under a binding internal rule that no valuation is claimable prior to MRR. Every figure below is code-verified: derived from running systems, passing test suites, or deployed artifacts. Claims that did not survive that audit were withdrawn and are listed in the withdrawn-claims register rather than quietly removed.

How to read this document

Written for investors, acquirers, and partners performing diligence. **Tier A** claims are code-verified: running code, passing tests, deployed artifacts, commit anchors. These are checkable. **Tier B** claims are built and demonstrable but not independently verified. Described, not asserted as proof. **Tier C** claims are specifications that do not execute. They are labelled as such and carry no value.

What this document deliberately does not contain. No valuation. No infrastructure detail: host addresses, project references, and credential state are withheld, not because they are unflattering, but because publishing them would be poor security practice and you should expect a counterparty to withhold them. No export-controlled content: three defense programs exist. They are named in Volume VII of the master codex and nothing more. Technical data on those programs is gated behind an ITAR/EAR review that has not been performed and will not be circumvented for a fundraise.

The honest summary, stated up front

One operator. No engineering hires. Roughly 43 working codebases, four deployed properties in July (three with no November re-verification, one now dark), live systems running inside real businesses, a parcel scoring model with 148 passing tests against 43,897 real records (July figures, not re-verified in November), and hash-chained provenance receipts on build claims.

MRR is $0. One product carries verified revenue. The flagship CRM is currently offline (deployment paused, November 4, 2026) and its production database is unreachable. The operation's demonstrated weakness is conversion, not construction. That paragraph is the whole thesis and its whole risk. Everything following is evidence for both halves.

Section 1. Verified capability inventory

Tier A only. Every number below survives adversarial diligence, or is labelled with the date it was last verified.

Figure	Value	As-of	Note
Real working codebases	about 43	Jul 2026	Tier 0+1+2 of 216 org repos (verified Nov 8, 2026)
Live / deployed properties	4 in July, 3 unverified in Nov, 1 dark	Nov 10, 2026	DeedSonar V3 dark (503) since Nov 4
Revenue-verified product	1 (TaxShield)	Jul 2026	No November re-verification on record
Sellable product lines	about 8	Jul 2026	See sale catalog in the master codex
Passing tests, landos-engine	148	Jul 2026	Not re-verified in November
	49 (Jul) / 50 (Nov)		Conflict, unresolved, both carried

Figure	Value	As-of	Note
Signals in the scoring model		Nov 10, 2026	
Real records scored	43,897	Jul 2026	Not re-verified in November
Named IP frameworks	31	Jul 2026	Count unchanged in November
MRR	$0	Nov 10, 2026	Binding rule: no valuation claimable pre-MRR

Section 2. Real estate intelligence stack

The most commercially mature vertical in the estate. Real data, real model, real tests, real delivery surface.

LandOS scoring substrate. A parcel-level scoring engine that ingests normalized county property records and emits a ranked acquisition score per parcel, using a 49-signal model (July) / 50-signal model (November estate record), conflict unresolved. Deterministic: the same input and model version always produce the same score. 148 passing tests (July). 43,897 real parcels scored (July). Two counties: St. Joseph IN and LaPorte IN. The scoring model is the moat; the CRM is the delivery vehicle. Licensable independently of DeedSonar.

landos-ingest. Live ETL pipeline pulling county property records into the normalized schema. Snapshot 1245b0b (July). Not independently saleable; it is the cost of entry for each new county.

Single-writer firewall. sync_to_crm.py is the sole permitted writer to the acquisition_leads table. Every scored lead is attributable to exactly one deterministic pipeline execution. Any mutation path that bypasses the gate is an integrity breach, not a convenience.

DeedSonar. White-label real-estate CRM and lead app. **Ownership corrected November 10, 2026: DeedSonar is solely Jeramie Hicks'. The July codex described a 50/50 joint venture; that framing is superseded. The intended partner never signed.** Current state (November 4, 2026): the site returns a deployment-paused error and the production database is unreachable. Not live as a running product today; offered as IP plus a buyer demo. The July production state (V3 alpha-final, merge 4674604) is historical.

DATUM. A valuation product beside LandOS, not a rebuild of it. The model emits facts; deterministic Python emits dollars. Indiana first (SDF Tier 0 resolved, July 2026), Michigan second. Repository version v0.4.0.

FALLOW. A standalone dual-engine land-yield venture: managed yield operations plus yield intelligence. Name locked, no entity filed, LaPorte pilot not run. Scoring engine v0.1.0 with 39 passing tests, verified November 10, 2026.

Section 3. Protocol, registry, and governance

ncell Protocol. TypeScript protocol library: 12 Zod schemas, 4 profiles. Wave 1 shipped (commit 39090e3, 952 insertions). Test count in conflict: 47 (July) vs 36/36 (November estate record). Waves 2-8 deliberately parked. Protocol libraries are worthless without adopters and extremely valuable with them; ncell currently has zero external consumers.

NRH Registry. Universal addressing system (NRH-[DD]-[ENT]-[TTT]-[NNN]) with a hash-chained transition log. 10 domains and 48 seeded entities (July) vs 9 domains and 35 seeded entities (November visual); discrepancy unresolved. Six open decisions block completion. No repository exists under the registry name in the organization; it lives as content, vault, and demos.

CRUCIBLE / BuildOS v2. A hash-chained receipt system acting as a publication gate. The production ledger (44 receipts) verifies as a valid chain under the current verifier, with every receipt hash recomputed and matching, 17 PASS / 27 FLAG against a 70.0 threshold. Keyed HMAC-SHA256 signing is implemented in the codebase, integrated into the service, and tested (34/34 assertions passed, November 10, 2026). Zero of the 44 production records carry a signature: no key has been generated or applied. The authorship gap is a deployment and custody gap, not a code gap. CRUCIBLE is the notary, not the judge: it proves sequence integrity, not authorship, until signing is applied.

Section 4. Agentic infrastructure

AXON Gateway. Phase 1 shipped April 1, 2026. **MOTHER daemon.** Monitoring and supervision stack. **Eight-stage autonomous nightly pipeline.** Unattended build loop; the mechanism that explains the output-to-headcount ratio, never packaged as a demonstration. **L4 Engine.** Ten-layer autonomous code synthesis: AST-surgical mutation, Darwinian synthesis, decision provenance with receipts. **NCP Phase 1.** 1,890 AtomicNodes and a claimed 98% LLM-bypass on the Semantic Cache Vault (July, not re-verified in November; the most economically interesting and least re-verified number in the tier). **LEGION.** Deterministic governed swarm: 6,487 opportunities scored in 3.3 seconds, 0 failed, sealed roll-up, merged September 30, 2026. A replay viewer is live.

Section 5. Revenue surface

Four assets were live in July; one (DeedSonar) is now dark, and the other three have no November re-verification in any record available to this codex. Five assets are built and earning nothing. Three are dark or blocked. One product carries all verified revenue. The build-to-monetize ratio is the defining problem of this operation, and it is visible in a single table. Every asset in the built column was harder to build than it would be to close.

Section 6. How close is he to doing the impossible?

Closer than the estate looks from the outside, and further than the build rate suggests from the inside. The impossible part was never building 43 codebases; one person with AI leverage can build 43 codebases. The impossible part is the last ten percent of each system: the key applied to the ledger, the database restored, the DNS swapped, the contract signed, the loop closed. By the measure that matters (wired, verified, earning), the estate is about 70 percent built, about 30 percent wired, and about 5 percent earning. The path from here is not another build. It is the knock-out list: key the ledger, restore the spine, swap the DNS, file the LLC, rule the kernel fork, close the seam. Six items. Every one of them is a decision followed by a command.

Appendix D. Withdrawn claims register

The following claims were previously made and have been formally withdrawn. They are published here rather than quietly deleted. A counterparty is entitled to know what was claimed before and what it was replaced with.

Withdrawn claim	Status	Replaced with
$6M indicative valuation	Withdrawn 30 Jun 2026	No dollar valuation is current or claimable pre-MRR
$750K to $1.75M	Withdrawn 30 Jun 2026	No dollar valuation is current or claimable pre-MRR
$15M to $25M pre-money	Withdrawn 30 Jun 2026	No dollar valuation is current or claimable pre-MRR
801 IP entries	Withdrawn 30 Jun 2026	31 named IP frameworks
858 assets	Withdrawn 30 Jun 2026	About 43 working codebases, 4 live properties, about 8 sellable lines
130 repos framed as 130 products	Withdrawn 30 Jun 2026	About 43 working codebases; org holds 216 repos (Nov 8, 2026)
Quantum-proof / Palantir-class capability	Withdrawn 30 Jun 2026	Withdrawn entirely. Not supportable.
62 passing tests on CRUCIBLE	Withdrawn 28 Jul 2026	No test files on disk in July. Superseded in part: repo now has three test files, 34/34 passed Nov 10, 2026.
CRUCIBLE HMAC-SHA256 ledger	Withdrawn 28 Jul 2026	Not implemented in July. Superseded in part: HMAC implemented, integrated, tested; production ledger still carries zero signed records.
CRUCIBLE is tamper-proof	Withdrawn 28 Jul 2026	Tamper-evident on sequence only.
		DeedSonar is solely Jeramie Hicks' (corrected Oct 22, 2026).

Withdrawn claim	Status	Replaced with
DeedSonar is a 50/50 joint venture	Withdrawn 10 Nov 2026	
DeedSonar V3 is live in production	Withdrawn 10 Nov 2026	Deployment paused; production database unreachable (Nov 4, 2026).
The organization holds 130 repos	Withdrawn 10 Nov 2026	216 repos (verified Nov 8, 2026).
prbench runs 7 models	Withdrawn 10 Nov 2026	Exactly 3 pinned Anthropic models; the 7-model config never landed.

Diagrams

Corrected diagram set, generated programmatically November 10, 2026. No diagram contains a host address, a project reference, a UUID, a credential, or a file path.

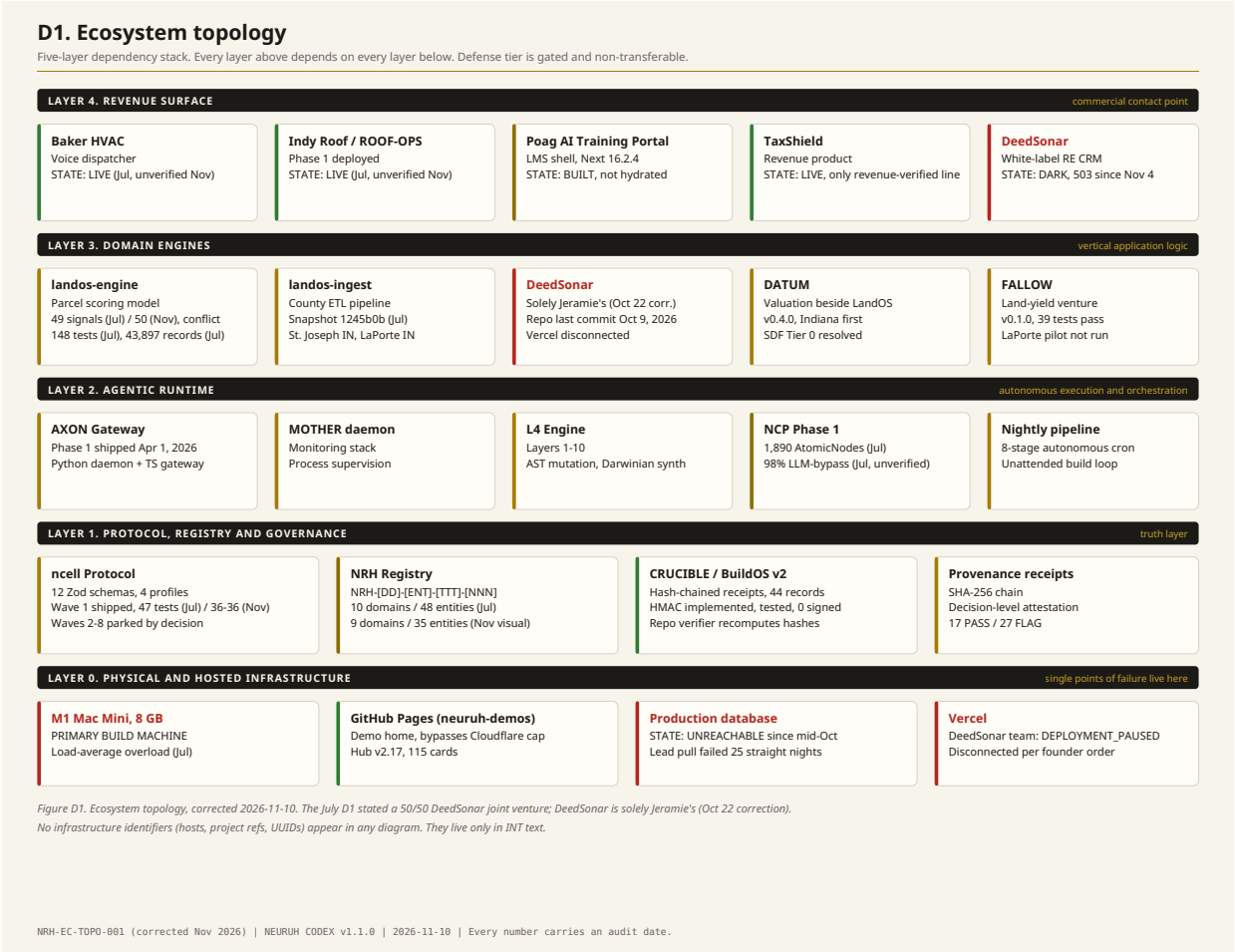


Figure D1. Ecosystem topology, corrected 2026-11-10.

D2. Real estate intelligence, data lineage

County record to closed disposition. The single-writer firewall is the integrity control for the entire vertical.

01. SOURCE
County records
St. Joseph IN
LaPorte IN

→

02. INGEST
landos-ingest
Live ETL
Snapshot 1245b0b (Jul)

→

03. NORMALIZE
Entity resolution
Parcel, owner
Address canonicalization

→

04. SCORE
landos-engine
49 signals (Jul)
50 signals (Nov), conflict

→

05. GATE
sync_to_crm.py
SINGLE WRITER
Integrity chokepoint

→

06. SERVE
DeedSonar CRM
Database + hosting
STATE: DARK, 503 Nov 4

SINGLE-WRITER FIREWALL
RULE: sync_to_crm.py is the sole writer to acquisition_leads.
WHY: multiple writers destroy score provenance.
EFFECT: any bypassing mutation path is an integrity breach,
not a convenience. No exceptions for UI, agent, or manual edit.
AUDIT: every write attributable to one pipeline run + model version.

SCORING SUBSTRATE, VERIFIED (JULY)
49 signals (Jul) / 50 signals (Nov estate record), conflict unresolved.
43,897 real parcels scored, not synthetic, not sampled.
148 passing tests on landos-engine (Jul, not re-verified Nov).
2 counties live: St. Joseph IN, LaPorte IN.
1 canonical repo: NeuruhAI/landos-engine (private).

DOWNSTREAM CONSUMERS AND TENANCY
DeedSonar (solely Jeramie's, Oct 22 correction) | Senior Assemblage Recipe (Lake Michigan shoreline, Tier B) |
Probate vertical (WA proposal delivered, TX warm, Tier C) | White-label channel (Turco track, config-separated resale) |
Migration 086 (917 lines, RLS rewrite, row-level tenancy) applied in July; July session found 084-086 not applied,
pipeline_stage_audit missing, promote_lead broken at runtime. Historical July state, not current.

Figure D2. Data lineage, corrected 2026-11-10. The July D2 stated a 50/50 DeedSonar joint venture; superseded.
November state: the CRM stage is dark (503) and the production database is unreachable. The firewall rule stands as architecture.

NRH-RE-LINEAGE-001 (corrected Nov 2026) | NEURUH CODEX v1.1.0 | 2026-11-10 | Every number carries an audit date.

Figure D2. Real estate data lineage and the single-writer firewall.

D4. CRUCIBLE / BuildOS v2, publication gate

The v1 defect and the v2 design that addresses it. State verified 2026-11-10 from the live repo.

BEFORE, BuildOS v1 (VULNERABLE)
Central defect: the model reads evidence it wrote.
Model asserts completion -> writes evidence to its own log
-> gate reads that same log.
CIRCULAR: the attestation loop closes on itself.
Consequence: a model that hallucinates completion also
hallucinates the proof of completion.

AFTER, BuildOS v2 (SEALED, WITH ONE GAP LEFT)
Completion gate joins claims against a ledger the model
cannot write. Guard rewritten with tokenizer-based detection.
VERIFIED: SHA-256 receipts, hash-chained, independently checkable.
VERIFIED 2026-11-10: 44 receipts, chain VALID, hashes recomputed.
REMAINING: 0 of 44 production records carry a signature.

WHAT THE REPO NOW PROVES (NOV 10)
HMAC-SHA256 implemented in receipt.py (generate/load/sign/verify).
Integrated into the service; signing.required=true makes a
missing key fatal. test_signing.py: 34/34 assertions passed.
Repo verifier recomputes every receipt_hash and catches
content edits (verified: tampered verdict fails, exit 1).

WHAT IS STILL MISSING (NOV 10)
No signing key has been generated or applied.
Production ledger: genesis epoch 44 records, signed epoch 0.
Key custody open: Mac key file, environment variable, or never made.
Uploaded verify_chain.py is the OLDER linkage-only version;
it does not detect content edits or tail truncation. Use the repo verifier.

WHY THIS IS THE MOST COMMERCIALLY IMPORTANT COMPONENT
Diligence: an acquirer can verify claims without trusting the founder's narrative. | Defensibility: verifiable autonomous build systems are not commoditised.
Regulatory: maps onto SOC2 audit-logging and ISO 27001 integrity controls. | Licensing: domain-agnostic, separable, licensable as standalone infrastructure.

Figure D4. CRUCIBLE publication gate. The July D4 stated keyed signing absent; Level 1 is now built, integrated, and tested, but unapplied.

NRH-GV-CRUCIBLE-001 (corrected Nov 2026) | NEURUH CODEX v1.1.0 | 2026-11-10 | Every number carries an audit date.

Figure D4. CRUCIBLE publication gate, state verified 2026-11-10.

D5. Revenue surface, state map (November 10, 2026)

Every commercial asset by state. Monetary figures excluded per operating rule; state and unblock action only.

LIVE, DEPLOYED AND RUNNING — proof the motion converts	BUILT, NOT YET EARNING — the near-term opportunity	DARK / BLOCKED — costs money, earns nothing
TaxShield Only verified revenue product. HOLD. Protect, do not refactor.	**Poag AI Training Portal** LMS shell ported. Build + audit passing. HYDRATE. Manus content in.	**DeedSonar V3** 503 DEPLOYMENT_PAUSED (Nov 4). DB dead. RESTORE SPINE, then redeploy.
Baker HVAC voice dispatcher Deployed in a real business. FORMALISE. Standing agreement.	**Senior Assemblage Recipe** Shoreline targeting overlay built. PACKAGE. Productise as a named offer.	**Bookeit** Built. Serving HTTP 402 (Jul). PAY. Vercel billing restoration.
Indy Roof, ROOF-OPS Phase 1 Phase 1 deployed and operating. EXPAND. Scope Phase 2 from usage.	**Valermus, Toronto** Partnership locked. Joint product imminent. SHIP. Define the v1 artifact.	**FSP / ResellOS** HTTP 402 + Firebase key rotation overdue. PAY + ROTATE, in that order.
	Min Qiu, WA probate v3 proposal delivered. CLOSE. Resolve transparency objection.	**DISPATCH-IQ** Scoped. No hardware. PROCURE. RTL-SDR confirmation.

NOVEMBER CORRECTION: the July D5 listed DeedSonar V3 as LIVE. It is dark (503) as of Nov 4, 2026.

TaxShield, Baker HVAC, and Indy Roof have no November re-verification in any record available to this codex; they are carried as last-known July state, not as current. Do not cite them as live without a fresh check.

Figure D5. Revenue surface state map. Column one proves conversion. Column two is the near-term opportunity. Column three costs money and earns nothing. The structural finding stands and sharpens: four assets were live in July; one of the four (DeedSonar) is now dark.

NRH-RV-STATE-001 (corrected Nov 2026) | NEURUH CODEX v1.1.0 | 2026-11-10 | Every number carries an audit date.

Figure D5. Revenue surface state map, November 10, 2026.

D7. FALLOW yield stack

Dual-engine land-yield venture. Engine A executes leases; Engine B sells the intelligence Engine A produces.

ENGINE A. MANAGED YIELD OPS — score, outbound, manage, match, collect

A1. MASTER LEASE (principal) Venture leases the parcel at fixed rent, sublets use-rights. Takes the whole spread and the rent risk. No broker license: principal position, not agency.	**A2. FLAT-FEE PLATFORM / SAAS** Owner subscribes to tooling, executes every lease itself. Venture never negotiates, never takes a cut of rent. No broker license: tooling, not brokerage.	**A3. BROKER-OF-RECORD AFFILIATION** Affiliate a licensed Michigan broker (and Indiana broker). Commission leasing runs through them. Week 4-8 conversation. A1 + A2 launch first.

ENGINE B. YIELD INTELLIGENCE — the intelligence layer, sold to everyone holding idle land

Land Yield Score SaaS seats for land investors. Per-parcel scoring API. Portfolio underwriting engagements.	**Demand heatmaps** Where laydown, storage, hunting, camping, apiary demand actually is. Data licensing.	**The moat: realized_yield** Every lease A executes becomes training data B sells. scoring_snapshot_id joins prediction to reality.

THE SCORE (v0.1.0, 39 TESTS PASSING) 42 signals + 7 hard gates. Gates suppress use verticals, not discounts. Every score carries score_version (lys-0.1.0), non-nullable. Stacking conflict matrix (C/S/T/X) with drafting conditions.	**LANE A PILOT (90 DAYS, NOT RUN)** LaPorte County. 500 mailers. Kill criteria at week 13 (any two fail): signed conversion below 2%; modeled-to-realized ratio below 0.6; demand-match below 50% within 60 days.

LICENSING CORRECTION: the 15-25% take-rate model is unlicensed brokerage in Michigan (MCL 339.2501, cited in thesis). Dead as specced.

LANE B: farm management firms and farmland funds (land banks and builders killed by the thesis's own research).

Figure D7. FALLOW yield stack. Name locked, no entity filed, LaPorte pilot not run. Live visual: neuruh-demos/fallow-yield/.

NRH-FL-STACK-001 (new Nov 2026) | NEURUH CODEX v1.1.0 | 2026-11-10 | Every number carries an audit date.

Figure D7. FALLOW yield stack.

D8. DATUM valuation DAG

The model emits facts. Deterministic Python emits dollars. The LLM structurally cannot emit a price.

LAYER 0.5 JURISDICTION RESOLVER
Returns an authority map, not a jurisdiction. Zoning, septic, drainage, floodplain, wetlands, assessment each resolve to a different body.

LAYER 0 SUBJECT RESOLUTION
Deterministic. Which parcel, which record, which authority, pinned by date.

LAYER 1 FACT EXTRACTION
The only genuinely parallel layer. Regulatory, infrastructure, physical, market pullers hit different sources.

LAYER 2 FEASIBILITY SOLVER
The only layer allowed to state what is buildable. Regulatory, infrastructure, soils resolve here. Double-counting dies here.

LAYER 3 SUBTYPE ROUTER
Selects the pricing engine.

LAYER 4 VALUATION ENGINES
Pure Python. No LLM arithmetic. Re-run last month's valuation from the cached feature vector, byte-identical.

LAYER 5 CALIBRATION
Versioned JSON coefficients, geography x subtype x YEAR, hash-stamped.

LAYER 6 RECONCILER
Fires only when engines disagree beyond a threshold. LLM judgment with the disagreement explicit, never averaged away.

LABEL TIERS
Tier 0: hard labels, recorded sales (SDF).
Tier 1: sales-ratio study (COD, PRD).
Tier 2: market signals (list, DOM, withdrawals).
Tier 3: own offer outcomes (censored, unreplicable).
Tier 4: founder judgment (structure, then decays).
Trust: Tier 0 drives the fit, Tier 3 eventually dominates.

THE LLM / PYTHON SEAM
Model emits: facts with evidence refs,
categorical judgments (closed enums),
ordinal judgments (bounded scales),
probabilities (0-1, separately calibrated),
abstention (null + evidence gap, first-class).
No dollar field exists in its output schema.

TIER 0, RESOLVED (JUL 25)
Indiana SDF: free download, UTF-16LE,
price /100, acreage /10000, per-SDF_ID.
St. Joseph 2024: 5,841 records, 2,262 Tier 0.
One $7.4B outlier quarantined (see II.6).

Figure D8. DATUM DAG. Loss: asymmetric pinball, overestimation penalized about 3:1 (config). Fit P10/P50/P90, not a point estimate.
Repo: NeuruhAI/datum, v0.4.0 (07184f5, Jul 26, 2026). Indiana first, Michigan second. Q-002 (offer history) still open.

NRH-DT-DAG-001 (new Nov 2026) | NEURUH CODEX v1.1.0 | 2026-11-10 | Every number carries an audit date.

Figure D8. DATUM valuation DAG.

D9. LEGION governed swarm, replay

10,000 governed workers on a shared thread pool. Deterministic replay with receipts, not swarm intelligence.

SUPERVISOR
One supervisor takes an exclusive lock.
HALT file checked before start and every dispatch.
Journal re-verified before resume; broken link halts.
Agents read only, cannot spawn, contact NOT_AUTHORIZED.

DISPATCH (THREAD POOL)
6,487 opportunities scored in 3.3 seconds.
0 failed. Tiers: A 208, B 1,452, C 4,793, gov-excluded 34.
Fault injection: 9,997 ok, 3 deliberate failures contained.
Crash resume produced byte-identical output.

SEALED ROLL-UP
50 results seal into a squad.
20 squads seal into a cohort.
7 cohorts seal into FINAL_OUTPUT.
Main ledger gets 11 records, not 6,487.

THE RUN (RECORDED 2026-09-28)
Run id legion-8b4fcca041981a3af08a6331. 130 squads, 7 cohorts.
387 tests pass, 25 of them LEGION's.
PR #26 merged 2026-09-30 (dbd9ef6).
Proof tag legion-v0.1.0-proof-20260928 at 8719d8d.

HONEST SCORECARD
Each agent is a deterministic scoring unit. No LLM in the loop.
10,000 identical scorers is parallel processing with receipts.
Rare parts: deterministic replay, sealed roll-up,
builder-separate-from-judge, limits enforced in code.

LIVE REPLAY
Viewer: neuruhai.github.io/neuruh-demos/legion-replay/ (34,116 B, 87 DOM checks, shipped 2026-11-10).
7 cohort clusters, 130 squad nodes, 6,487 tier-shaped agents, sealed FINAL_OUTPUT core. Layout derived, not recorded telemetry.
LEGION-LIVE (:8620/:8621) is specified, not built. Until then, any live swarm visual is a replay and must be labeled as one.

Figure D9. LEGION replay. LEGION is a job, not a service. It runs for seconds and exits.

NRH-LG-REPLAY-001 (new Nov 2026) | NEURUH CODEX v1.1.0 | 2026-11-10 | Every number carries an audit date.

Figure D9. LEGION governed swarm, replay.